

# CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

## PRE-MEDICAL : ENTHUSIAST COURSE [ PHASE-ALL ]

**TEST # 02**
**DATE : 25-04-2020**
**Test Type : Minor**
**Test Pattern : NEET**

### INSTRUCTIONS

**Do not open this Test Booklet until you are asked to do so**

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry and Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

**ALLEN** CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,  
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi  
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

www.allen.ac.in

**Your Target is to secure Good Rank in NEET 2020**

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. The greatest value of the function  $-5 \sin \theta + 12 \cos \theta$  is :-  
(1) 12 (2) 13 (3) 7 (4) 17

2. Two point charges  $+9q$  and  $+q$  are kept 16 cm apart. Where should a third charge  $q$  be placed between them so that the system remains in equilibrium ?

- (1) 24 cm from  $+9q$  (2) 12 cm from  $+9q$   
(3) 24 cm from  $+q$  (4) 12 cm from  $+q$

3. Two small spheres each having a charge  $+Q$  are suspended by insulating threads of length  $L$  from a hook. This arrangement is taken to a space where there is no gravitational effect, then the angle between the two threads and the tension in each will be :-

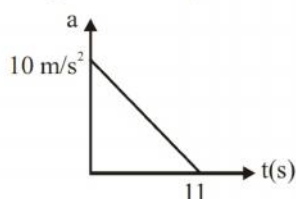
- (1)  $180^\circ, \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(2L)^2}$  (2)  $90^\circ, \frac{1}{4\pi\epsilon_0} \frac{Q^2}{L^2}$   
(3)  $180^\circ, \frac{1}{4\pi\epsilon_0} \frac{Q^2}{2L^2}$  (4)  $180^\circ, \frac{1}{4\pi\epsilon_0} \frac{Q^2}{L^2}$

4. The equation of the stationary wave is :-

$$y = 2A \sin \left[ \frac{2\pi ct}{\lambda} \right] \cos \left[ \frac{2\pi x}{\lambda} \right]$$

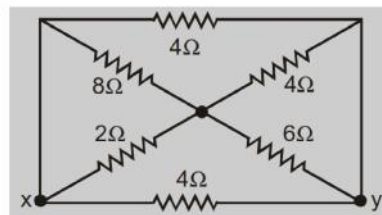
Which of the following statements is wrong?

- (1) The unit of  $ct$  is same as that of  $\lambda$   
(2) The unit of  $x$  is same as that of  $\lambda$   
(3) The unit of  $2\pi c/\lambda$  is same as that of  $2\pi x/\lambda t$   
(4) The unit of  $c/\lambda$  is same as that of  $x/\lambda$
5. A particle starts from rest. Its acceleration ( $a$ ) versus time ( $t$ ) is as shown in the figure. The maximum speed of the particle will be :-



- (1) 110 m/s (2) 55 m/s  
(3) 550 m/s (4) 660 m/s

6. The total resistance between  $x$  and  $y$  in ohms is :-



- (1)  $1 \Omega$  (2)  $4 \Omega$  (3)  $\frac{4}{3} \Omega$  (4)  $\frac{2}{3} \Omega$

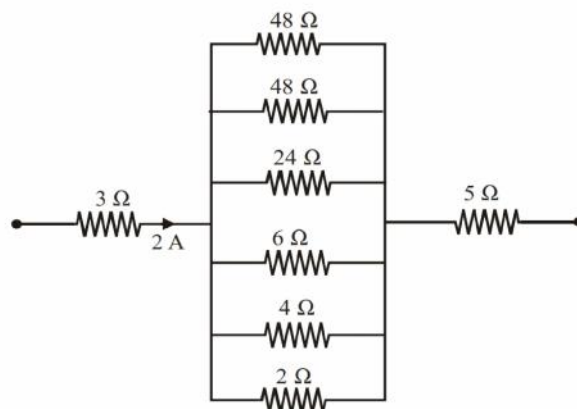
7. A particle is projected with a velocity  $v$ , so that its range on a horizontal plane is twice the greatest height attained. If  $g$  is acceleration due to gravity, then its range is :-

- (1)  $\frac{4v^2}{5g}$  (2)  $\frac{4g}{5v^2}$  (3)  $\frac{4v^3}{5g^2}$  (4)  $\frac{4v}{5g^2}$

8. A particle is fired with velocity  $u$  making an angle  $\theta$  with the horizontal. What is the magnitude of change in velocity when it is at the highest point ?

- (1)  $u \cos \theta$  (2)  $u$   
(3)  $u \sin \theta$  (4)  $(u \cos \theta - u)$

9. Find the potential difference across the  $24 \Omega$  :-

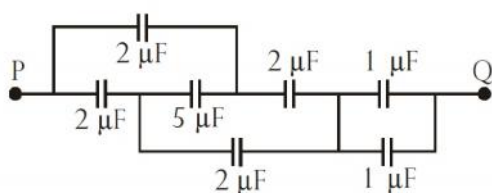


- (1) 48 volts (2) 2 volts  
(3) 4 volts (4) 1 volts

10. If the net force on a body is zero, will it definitely be at rest :-

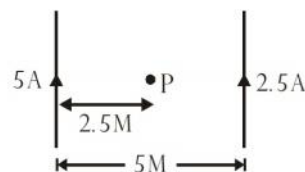
- (1) Yes  
(2) Not necessary. This is possible that the body is moving with accelerating velocity  
(3) Not necessary. This is possible that the body is moving with decelerating velocity  
(4) Not necessary. This is possible that the body is moving with constant velocity

11. The effective capacitance between the points P and Q of the arrangement shown in the figure is



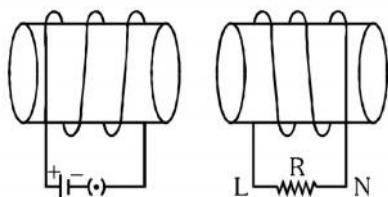
- (1)  $(1/2) \mu\text{F}$  (2)  $1 \mu\text{F}$   
(3)  $2 \mu\text{F}$  (4)  $1.33 \mu\text{F}$
12. A reference frame attached to the earth :-  
(1) is an inertial frame because Newton's laws of motion are applicable in this frame  
(2) is an inertial frame by definition  
(3) cannot be an inertial frame because earth is revolving round the sun  
(4) None of these
13. A car starts from rest with a constant acceleration of  $2 \text{ m/s}^2$ . After 5 s, a ball is dropped through the window of the car. The window of the car is at a height of 1.5 m from the ground. What will be the speed of the ball 0.5 s after it was dropped? (Take  $g = 10 \text{ m/s}^2$ ):-  
(1) 5 m/s (2)  $\sqrt{5} \text{ m/s}$   
(3)  $5\sqrt{5} \text{ m/s}$  (4) 2 m/s
14. A capacitor of capacitance  $5 \mu\text{F}$  is connected as shown in the figure. The internal resistance of the cell is  $0.5 \Omega$ . The amount of charge on the capacitor plate is :  
(1) 0  
(2)  $5 \mu\text{C}$   
(3)  $10 \mu\text{C}$   
(4)  $25 \mu\text{C}$
- 
15. When a body moves uniformly in a circular path, no work is done by the force since :-  
(1) there is not net force  
(2) there is no displacement  
(3) force is always away from the centre  
(4) force and displacement are perpendicular to each other

16. For the given current distribution the magnetic field at point, 'P' is :-



- (1)  $\frac{\mu_0}{4\pi} \odot$  (2)  $\frac{\mu_0}{\pi} \otimes$  (3)  $\frac{\mu_0}{2\pi} \otimes$  (4)  $\frac{\mu_0}{2\pi} \odot$
17. A particle covers equal distance around a circular path, in equal intervals of time. Which of the following quantities connected with the motion of the particle remains constant with time ?  
(1) Displacement (2) Velocity  
(3) Speed (4) Acceleration
18. A particle is moving along a circular path. The angular velocity, linear velocity, angular acceleration and centripetal acceleration of the particle at any instant respectively are  $\vec{\omega}, \vec{v}, \vec{\alpha}$  and  $\vec{a}_c$ . Which of the following relations is not correct ?  
(1)  $\vec{\omega} \perp \vec{v}$  (2)  $\vec{\omega} \perp \vec{\alpha}$   
(3)  $\vec{\omega} \perp \vec{a}_c$  (4)  $\vec{v} \perp \vec{a}_c$
19. A closed loop PQRS carrying a current is placed in a uniform magnetic field. If the magnetic forces on segments PS, SR and RQ are  $F_1, F_2$  and  $F_3$  respectively and are in the plane of the paper and along the directions shown, the force on the segment QP is :-  
(1)  $\sqrt{(F_3 - F_1)^2 - F_2^2}$   
(2)  $F_3 - F_1 - F_2$   
(3)  $F_3 - F_1 - F_2$   
(4)  $\sqrt{(F_3 - F_1)^2 + F_2^2}$
- 
20. A bullet weighing 50 gm leaves the gun with a velocity of 30 m/s. If the recoil speed imparted to the gun is 1 m/s, the mass of the gun is:-  
(1) 15 kg (2) 30 kg  
(3) 1.5 kg (4) 20 kg
21. The centre of mass of a system of particle does not depend on:-  
(1) masses of the particles  
(2) forces acting on the particles  
(3) position of the particles  
(4) All of these

22. Two co-axial solenoids shown in figure. If key of primary suddenly opened then direction of instantaneous induced current in resistance 'R' which connected in secondary:

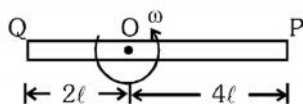


- (1) L to N (2) N to L  
(3) Alternating (4) Zero

23. A uniform disc of mass  $M$  and radius  $R$  is mounted on an axle supported in frictionless bearings. A light cord is wrapped around the rim of the disc and a steady downward pull  $T$  is exerted on the cord. The angular acceleration of the disc is:-

- (1)  $T/MR$  (2)  $MR/T$   
(3)  $2T/MR$  (4)  $MR/2T$

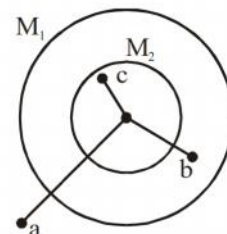
24. A conducting rod rotates with a constant angular velocity ' $\omega$ ' about the axis which passes through point 'O' and perpendicular to its length. A uniform magnetic field ' $B$ ' exists parallel to the axis of the rotation. Then potential difference between the two ends of the rod is :



- (1)  $6B\omega\ell^2$  (2)  $B\omega\ell^2$  (3)  $10B\omega\ell^2$  (4) Zero

25. A spherical solid ball of 1 kg mass and radius 3 cm is rotating about an axis passing through its centre with an angular velocity of 50 radian/sec. The kinetic energy of rotation is :-  
(1) 4500 J (2) 90 J  
(3)  $(9/20)$  J (4)  $(9/10)$  J

26. Two concentric spherical shells of uniform density of mass  $M_1$  and  $M_2$  are situated as shown in figure. The force on a particle of mass  $m$  located at the position  $r = a$  (The distance  $r$  is measured from the centre of the shells) is:-

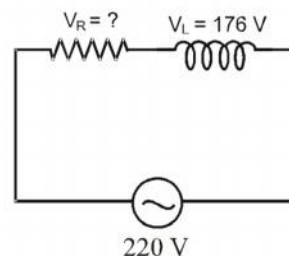


- (1)  $\frac{GM_1m}{b^2}$   
(2)  $\frac{GM_2m}{b^2}$   
(3)  $\frac{G(M_1 + M_2)m}{a^2}$   
(4) zero

27. In Q. no. 26, the force on the particle of mass  $m$ , when it is located at  $r = b$  is:-

- (1)  $\frac{GM_1m}{b^2}$  (2)  $\frac{GM_2m}{b^2}$   
(3)  $\frac{G(M_1 + M_2)m}{a^2}$  (4) zero

28. In showing figure find  $V_R$  :

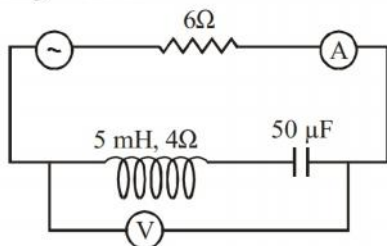


- (1) 132V (2) 396V  
(3) 185 V (4)  $\sqrt{220 \times 176}$

29. A simple harmonic motion has an amplitude  $A$  and time period  $T$ . The time required by it travel from  $x = A$  to  $x = A/2$  is :-

- (1)  $\frac{T}{12}$  (2)  $\frac{T}{4}$   
(3)  $\frac{T}{6}$  (4)  $\frac{T}{2}$

30. In the circuit shown in the figure, the A.C. source gives a voltage  $V = 20 \cos(2000t)$  volt neglecting source resistance, the voltmeter and ammeter readings will be :



- (1) 0V, 1.4A (2) 5.6V, 1.4A  
(3) 0V, 0.47 A (4) 1.68 V, 0.47 A

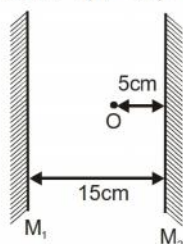
31. For an EM wave,

$$E = E_0 \sin 12 \times 10^6 [Z - 2 \times 10^8 t] \frac{N}{C}$$

in a medium, then its refractive index is

- (1) 2/3 (2) 3/2  
(3) 4/3 (4) 5/3
32. 'Stationary waves' are so called because in them :-  
(1) there occurs no flow of energy along the wave  
(2) the particles of the medium do not execute SHM  
(3) the particles of the medium are not disturbed at all  
(4) the interference effect cannot be observed
33. Two identical piano wires have a fundamental frequency of 600 cycle per second when kept under the same tension. What fractional increase in the tension of one wire will lead to the occurrence of 6 beats per second when both wires vibrate simultaneously:-  
(1) 0.01 (2) 0.02  
(3) 0.03 (4) 0.04

34. Figure shows two plane mirrors parallel to each other and an object O placed between them. Then the distance of the first three images from the mirror  $M_2$  will be :(in cm)



- (1) 5, 10, 15 (2) 5, 15, 30  
(3) 5, 25, 35 (4) 5, 15, 25

35. A convex lens is made up of three different materials as shown in the figure. For a point object placed on its axis, the number of images formed are



- (1) 1 (2) 3 (3) 4 (4) 5

36. When a thin transparent plate of thickness  $t$  and refractive index  $\mu$  is placed in the path of one of the two interfering waves of light, then the path difference changes by :

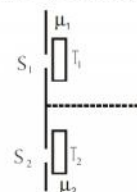
- (1)  $(\mu + 1)t$  (2)  $(\mu - 1)t$   
(3)  $\frac{(\mu + 1)}{t}$  (4)  $\frac{(\mu - 1)}{t}$

37. A semiconducting device is connected in a series circuit with a battery and a resistance. A current is found to pass through the circuit. If the polarity of the battery is reversed, the current drops to almost zero. The device may be:-

- (1) a p-n junction  
(2) an intrinsic semiconductor  
(3) a p-type semiconductor  
(4) an n-type semiconductor

38. In a Young's double slit experiment, a slab of thickness  $1.2 \mu\text{m}$  and refractive index 1.5 is placed in front of one slit and another slab of thickness  $t$  and refractive index 2.5 is placed in front of the second slit. If the position of the central fringe remains unaltered, then the thickness  $t$  is-

- (1)  $0.4 \mu\text{m}$   
(2)  $0.8 \mu\text{m}$   
(3)  $1.2 \mu\text{m}$   
(4)  $7 \mu\text{m}$

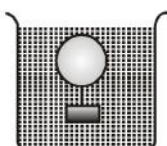


39. Out of the common-base, common-emitter and common-collector configurations of the transistor amplifier, the voltage gain is the highest in:-

- (1) common-base  
(2) common-collector  
(3) common-emitter  
(4) equal in all the cases

40. The magnitude of De Broglie wavelength ( $\lambda$ ) of electron (e), proton (p), neutron (n) and  $\alpha$ -particle ( $\alpha$ ) all having the same kinetic energy of 1 MeV, in the increasing order will follow the sequence:

- (1)  $\lambda_e, \lambda_p, \lambda_n, \lambda_\alpha$  (2)  $\lambda_e, \lambda_n, \lambda_p, \lambda_\alpha$   
(3)  $\lambda_\alpha, \lambda_n, \lambda_p, \lambda_e$  (4)  $\lambda_p, \lambda_e, \lambda_\alpha, \lambda_n$

41. Specific heat of a gas undergoing adiabatic change is:-  
 (1) zero (2) infinite  
 (3) positive (4) negative
42. Two separate monochromatic light beams A and B of the same intensity are falling normally on a unit area of a metallic surface. Their wavelength are  $\lambda_A$  and  $\lambda_B$  respectively. Assuming that all the incident light is used in ejecting the photoelectrons, the ratio of the number of photoelectrons from the beam A to that from B is  
 (1)  $(\lambda_A/\lambda_B)^2$  (2)  $\lambda_A/\lambda_B$   
 (3)  $\lambda_B/\lambda_A$  (4) 1
43. Mean kinetic energy per gram molecule for diatomic gas is:-  
 (1)  $\frac{3}{2}RT$  (2)  $\frac{4}{2}RT$   
 (3)  $\frac{5}{2}RT$  (4)  $\frac{6}{2}RT$
44. A body floats in a liquid contained in a beaker. The whole system shown in figure is falling under gravity. The upthrust on the body due to liquid is:-  
  
 (1) zero  
 (2) equal to weight of liquid displaced  
 (3) equal to weight of the body in air  
 (4) equal to weight of the immersed body
45. A certain mass of Hydrogen is changed to Helium by the process of fusion. The mass defect in fusion reaction is 0.02866 u. The energy liberated per u is : (given 1u = 931 MeV)  
 (1) 13.35 MeV (2) 2.67 MeV  
 (3) 26.7 MeV (4) 6.675 MeV

46. On reacting calgon with  $MgSO_4$ , the product obtained has a chemical formula :  
 (1)  $Na_2[Na_4(PO_3)_6]$  (2)  $Na_2[Mg_2(PO_4)_6]$   
 (3)  $Na_2[Na_4(PO_4)_6]$  (4)  $Na_2[Mg_2(PO_3)_6]$
47. Correct order of thermal stability :  
 (1)  $MgCO_3 > BeCO_3 > CaCO_3 > SrCO_3$   
 (2)  $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3$   
 (3)  $SrCO_3 > CaCO_3 > MgCO_3 > BeCO_3$   
 (4) None of these
48. Which of the following complex shows geometrical isomerism :  
 (1)  $[Zn(NH_3)_2Cl_2]$  (2)  $[Rh(PPh_3)_3Cl]$   
 (3)  $[Pd(NH_3)_2ClBr]$  (4)  $[Pt(en)ClBr]$
49. Total number of coordination isomers shown by complex  $[Co(NH_3)_6][CrCl_6]$  is :  
 (1) 4 (2) 5 (3) 6 (4) 0
50. Which of the following complex absorb visible light :  
 (1)  $[Cu(NH_3)_6]Cl_3$  (2)  $[Zn(NH_3)_6]Cl_2$   
 (3)  $[Ti(en)_2(SO_4)_2]$  (4)  $[Sc(H_2O)_3(NH_3)_3]Cl_3$
51. Hybridisation of  $I_3^-$  ion :  
 (1)  $sp^2$  (2)  $sp$   
 (3)  $sp^3d^2$  (4)  $sp^3d$
52. Which of the following species is paramagnetic:  
 (1)  $O_2$  (2)  $N_2$  (3)  $O_2^{2-}$  (4)  $F_2$
53. Which of the following anion has the smallest radius ?  
 (1)  $H^-$  (2)  $F^-$  (3)  $Cl^-$  (4)  $Br^-$
54. Element having highest I.P. value is :  
 (1) Ne (2) He  
 (3) Be (4) N
55.  $[X] + H_2SO_4 \rightarrow [Y]$  ; a colourless gas with irritating smell :  
 $[Y] + K_2Cr_2O_7 + H_2SO_4 \rightarrow$  Green solution  
 [X] and [Y] are :  
 (1)  $SO_3^{2-}$ ,  $SO_2$  (2)  $Cl^-$ ,  $HCl$   
 (3)  $S^{2-}$ ,  $H_2S$  (4)  $CO_3^{2-}$ ,  $CO_2$
56. Water is oxidised to oxygen by :  
 (1)  $ClO_2$  (2)  $KMnO_4$   
 (3)  $H_2O_2$  (4) Fluorine
57. Which of the following is the increasing order of enthalpy of vapourization ?  
 (1)  $NH_3 < PH_3 < AsH_3$   
 (2)  $AsH_3 < PH_3 < NH_3$   
 (3)  $NH_3 < AsH_3 < PH_3$   
 (4)  $PH_3 < AsH_3 < NH_3$

58. Brine solution on electrolysis will not give :

- (1) NaOH (2) O<sub>2</sub>  
(3) Cl<sub>2</sub> (4) H<sub>2</sub>

59. Which metallurgy not involves leaching :

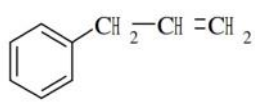
- (1) Au (2) Ag  
(3) Al (4) Fe

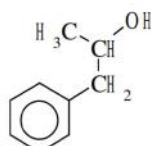
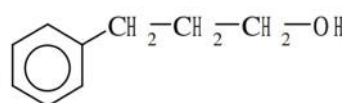
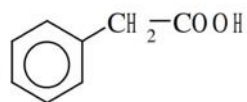
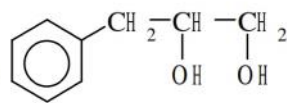
60. CaC<sub>2</sub> contain only :

- (1) Coordinate bond and Ionic bond  
(2) Covalent bond and coordinate bond  
(3) Both of the above  
(4) None of the above

61. The solubility of iodine in H<sub>2</sub>O may be increased by the addition of :

- (1) Na<sub>2</sub>S<sub>2</sub>O<sub>3</sub>  
(2) CHCl<sub>3</sub> and CCl<sub>4</sub>  
(3) KI  
(4) CS<sub>2</sub> and inorganic benzene

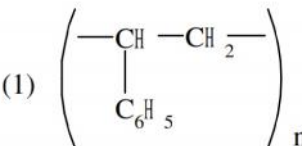
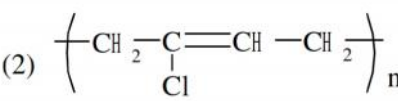
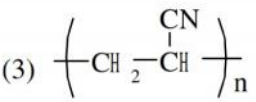
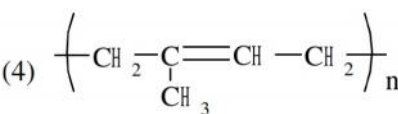
62.  on oxymercuration-demercuration produces the major product

- (1)   
(2)   
(3)   
(4) 

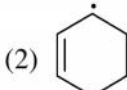
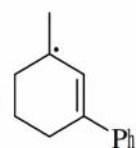
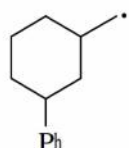
63. 1.4 g of an organic compound was digested according to Kjeldahl's method and the ammonia evolved was absorbed in 60 mL of M/10 H<sub>2</sub>SO<sub>4</sub> solution. The excess sulphuric acid requires 20 mL of M/10 NaOH solution for neutralisation. The percentage of nitrogen in the compound is

- (1) 3 (2) 5  
(3) 24 (4) 10

64. Which of the following structures represents Neoprene polymer

- (1)   
(2)   
(3)   
(4) 

65. Which one of the following is most stable

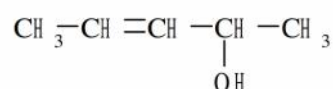
- (1)  (2)   
(3)  (4) 

66.  $\text{CH}_3\text{CH}_2\text{COOH} \xrightarrow{\text{SOCl}_2} \text{A} \xrightarrow[\Delta]{\text{NH}_3} \text{B} \xrightarrow[\text{KOH}]{\text{Br}_2} \text{C}$

Structure of compound C is

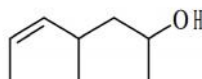
- (1) CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>NH<sub>2</sub> (2) CH<sub>3</sub>CH<sub>2</sub>NH<sub>2</sub>  
(3) CH<sub>3</sub>CH<sub>2</sub>NHCH<sub>3</sub> (4) CH<sub>3</sub>CH<sub>2</sub>CONH<sub>2</sub>

67. Number of stereoisomers of the given compound is



- (1) 2 (2) 3 (3) 4 (4) 6

68. IUPAC name of given compound is



- (1) 1,3-Dimethylhex-4-en-1-ol
- (2) 4,6-Dimethylhex-2-en-6-ol
- (3) 4-Methylhept-2-en-6-ol
- (4) 4-methylhept-5-en-2-ol

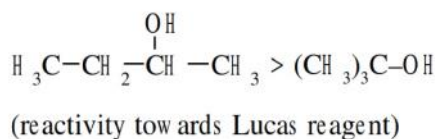
69. Which of the following is correct for

- (1)  $\text{CH}_3\text{-COCl} > \text{CH}_3\text{-COOCCH}_3 > \text{CH}_3\text{-COOC}_2\text{H}_5 > \text{CH}_3\text{-CONH}_2$  (reactivity towards hydrolysis)

- (2)  $\text{C}_6\text{H}_5\text{-CH=O} > \text{CH}_3\text{-CH=O} > \text{CH}_3\text{-CO-CH}_3$  (reactivity towards cyanohydrine formation)

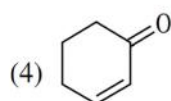
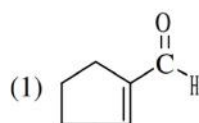
- (3)  $>$   $>$   $>$  (Reactivity towards electrophilic substitution reaction)

- (4)  $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH} >$

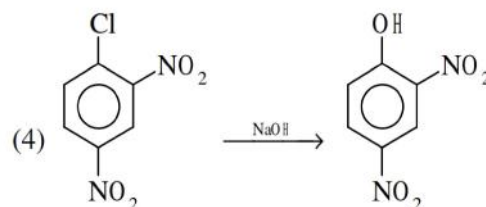
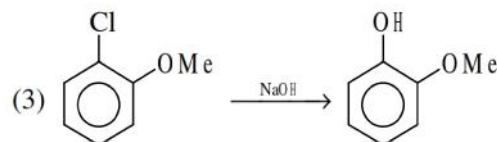
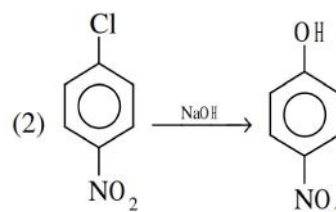
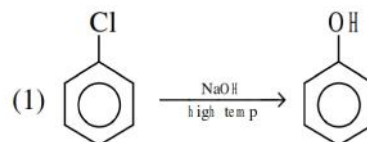


70.  $\xrightarrow[\text{H}_2\text{O}]{\text{O}_3}$  (A)  $\xrightarrow[\text{(ii) } \Delta]{\text{(i) Ca(OH)}_2}$  (B)

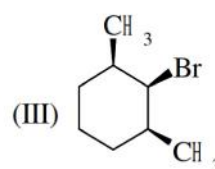
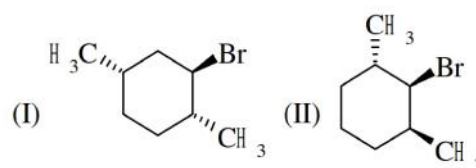
Product B is



71. Which reaction is least feasible



72. Arrange the following towards their reactivity for  $\text{E}^2$  elimination?



- (1) III > I > II
- (2) I > II > III
- (3) II > I > III
- (4) III > II > I

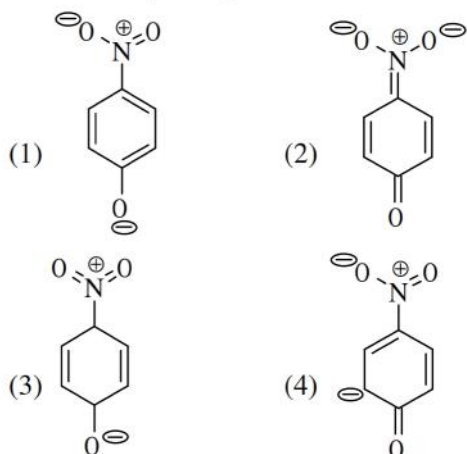
73.  $\text{CH}_3\text{CHO} \xrightarrow{\text{NaOH}}$  major product

- (1)  $\text{CH}_3\text{CH}_2\text{-O-CH}_2\text{-CH}_3$
- (2)  $\text{COONa}$
- (3)  $\text{CH}_3\text{CH}_2\text{-O-CH}_2\text{-COONa}$
- (4)  $\text{CH}_3\text{CH}_2\text{-O-CH}_2\text{-CH}_2\text{-O-CH}_2\text{-CH}_3$

74. Which of the following is not a monosaccharide?

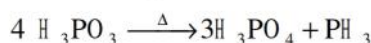
- (1) Glucose
- (2) Fructose
- (3) Cellulose
- (4) D-Ribose

75. The most unlikely representation of resonance structure of p-nitrophenoxide is



76. If first ionisation potential of an atom is 36 V then the second excitation potential will be :-  
(1) 10.2 V (2) 12 V  
(3) 27 V (4) 32 V

77. What will be the equivalent weight of  $\text{H}_3\text{PO}_3$  in the following reaction.



- (1)  $\frac{M}{6}$  (2)  $\frac{2M}{3}$  (3)  $\frac{M}{12}$  (4)  $\frac{M}{2}$

78. Reaction  $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$  is used to produce ammonia. When 450 g of hydrogen is reacted with nitrogen then 1575 g ammonia is produced. What is the percentage yield of this reaction :-

- (1) 41.5% (2) 30.8% (3) 61.8% (4) 20.7%

79. Under similar conditions which of the following gases will diffuse four times as quickly as oxygen ?

- (1) He (2)  $\text{H}_2$  (3)  $\text{N}_2$  (4)  $\text{D}_2$

80. An element (atomic mass = 250) crystallises in simple cubic lattice structure. If the density of unit cell is  $7.2 \text{ g cm}^{-3}$  then the atomic radius of the element will be :-

- (1)  $1.93 \times 10^{-6} \text{ cm}$  (2)  $1.93 \times 10^{-8} \text{ cm}$   
(3)  $1.93 \times 10^{-8} \text{ \AA}$  (4)  $1.93 \times 10^{-7} \text{ m}$

81. 20 g of naphthoic acid ( $\text{C}_{11}\text{H}_8\text{O}_2$ ) is dissolved in 50g of benzene then freezing point depression of 2K is observed for solution. The vant h off factor will be ( $K_f$  for benzene =  $1.72 \text{ K g mol}^{-1}$ )

- (1) 0.75 (2) 2 (3) 0.5 (4) 3

82. The highest pH value is of :-

- (1) 0.1 M NaCl (2) 0.1 M  $\text{NH}_4\text{Cl}$   
(3) 0.1 M  $\text{CH}_3\text{COONa}$  (4) 0.1 M  $\text{CH}_3\text{COONH}_4$

83. For the equilibrium  $\text{N}_2\text{O}_{4(g)} \rightleftharpoons 2\text{NO}_{2(g)}$  the ratio of theoretical and experimental vapour density is 1.5 then percentage amount of  $\text{NO}_2$  by volume in equilibrium mixture is :-

- (1) 50% (2) 100% (3) 33% (4) 67%

84. If a gas absorbs 200 J of heat and expands by  $500 \text{ cm}^3$  against a constant pressure  $2 \times 10^5 \text{ Nm}^{-2}$  then change in internal energy is :-

- (1) - 300 J (2) - 100 J (3) + 100 J (4) + 300 J

85. Incorrect statement is :-

- (1) In certain conditions the process with decrease in entropy may be spontaneous.  
(2) The entropy of a perfectly crystalline solid is zero at absolute zero temperature.  
(3) Enthalpy of combustion is always negative.  
(4) The process showing increase in free energy is spontaneous.

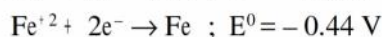
86. Henderson equation  $\text{pH} - \text{pK}_a = 5$  will be applicable to an acidic buffer when :-

- (1)  $[\text{Acid}] = [\text{conjugate base}]$   
(2)  $[\text{Acid}] \times 10^5 = [\text{conjugate base}]$   
(3)  $[\text{Acid}] = [\text{conjugate base}] \times 10^5$   
(4)  $[\text{Acid}] = 2 [\text{conjugate base}]$

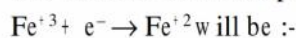
87. The rate law for reaction  $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$  is - rate =  $K [\text{A}]^{1/2} [\text{B}]^{1/2}$ . If the reaction starts from 1M initial concentration for both A and B then the time taken for concentration of A to be reduced to 0.25 M will be - ( $K = 2.303 \times 10^{-3} \text{ sec}^{-1}$ )

- (1) 300 sec (2) 600 sec  
(3) 900 sec (4) 1200 sec

88.  $\text{Fe}^{+3} + 3\text{e}^- \rightarrow \text{Fe}$  ;  $E^0 = -0.036 \text{ V}$



The standard electrode potential for



- (1) -0.476 V (2) -0.404 V  
(3) 0.44 V (4) 0.772 V

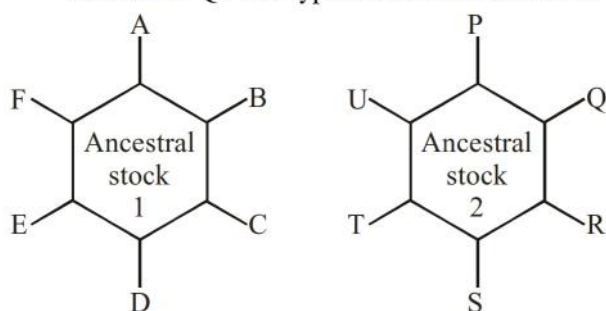
89. The most effective coagulating agent for antimony sulphide ( $\text{Sb}_2\text{S}_3$ ) sol is :-

- (1)  $\text{Na}_2\text{SO}_4$  (2)  $\text{CaCl}_2$   
(3)  $\text{Al}_2(\text{SO}_4)_3$  (4)  $\text{NH}_4\text{Cl}$

90. 3 moles of electrons are passed through  $\text{AgNO}_3$ ,  $\text{CuSO}_4$  and  $\text{AuCl}_3$  solutions connected in series then the molar ratio of Ag, Cu and Au deposited on their respective cathodes will be :-

- (1) 1 : 2 : 3 (2) 1 : 3 : 6  
(3) 6 : 3 : 1 (4) 6 : 3 : 2

91. Two adaptive radiations are shown in the figure below. Animal 'B' acquires the same adaptations as the Animal 'Q'. This type of evolution is called as :-



- (1) Convergent evolution  
(2) Divergent evolution  
(3) Both Convergent and divergent  
(4) Retrogressive evolution
92. Read two statements (a) and (b) carefully and select the **correct** option regarding them :-  
(a) First organisms appeared on earth were chemohautotrophic and anaerobic.  
(b) Cyanobacteria were the first organisms which liberated the oxygen in atmosphere.

**Options :**

- (1) Statement (a) is correct but statement (b) is incorrect  
(2) Statement (a) is incorrect but statement (b) is correct  
(3) Both statements are correct  
(4) Both statements are incorrect
93. Which was the first human being like creature ?  
(1) *Ramapithecus* (2) *Dryopithecus*  
(3) *Australopithecines* (4) *Homo habilis*
94. Elephantiasis is a chronic inflammation that results in gross deformities is caused by :-  
(1) *Trichophyton* (2) *E.coli*  
(3) *Wuchereria* (4) *Ascaris*
95. The second line of defence in the immune system is provided by ?  
(1) skin & mucous membrane  
(2) fever & pH  
(3) interferons & macrophage  
(4) 2 & 3 both
96. Viral DNA after being converted from viral RNA by 'X' enzyme, incorporates into host genome to undergo replication, what is 'X' ?  
(1) DNA polymerase  
(2) Restriction endonuclease  
(3) RNA polymerase  
(4) Reverse transcriptase

97. Which compound is formed by acetylation of morphine ?  
(1) Heroin (2) Cocaine  
(3) Tobacco (4) Marijuana
98. Which one at the following stage is **correctly** described with its event.

|     |            |                                                        |
|-----|------------|--------------------------------------------------------|
| (1) | Anaphase I | The bivalent chromosomes align on the equatorial plate |
| (2) | Diakinesis | Beginning of dissolution of synaptonemal complex       |
| (3) | Metaphase  | Spindle fibres attach to kinetochore of chromosomes    |
| (4) | Zygotene   | Appearance of recombination nodule                     |

99. The figure given below figure represent the stage of cell division. Read the following statement.



- (i) Nucleolus, Golgi complex and ER reform  
(ii) Chromosomes move to opposite pole  
(iii) Activity of recombinase enzyme  
(iv) Homologous chromosomes separate while sister chromatids associated at their centromere  
(v) Initiation of the assembly of mitotic spindle.
- How many of the above statement are **not true** with respect to given figure.
- (1) Four (2) Three  
(3) Five (4) Two

100. Every chromosome essentially has the A on the sides of which disc shaped structures called B are present.  
Choose the **correct** words for the blanks A and B respectively.
- (1) Centrosome, centrioles
  - (2) Kinetochore, centrosome
  - (3) Centromere, kinetochores
  - (4) DNA, RNA
101. A scientist made an observation regarding the protein synthesis. He found that ribosome formation take place in nucleolus. Which of the following is true for a cell that is actively carrying out protein synthesis. This cell contains :-
- (1) Larger and less numerous nucleoli
  - (2) Smaller and more numerous nucleoli
  - (3) Large and more numerous nucleoli
  - (4) Smaller and less numerous nucleoli
102. The arrangement of axonemal microtubules in cilia is referred to as the :-
- (1) 9 + 2 array
  - (2) 0 + 2 array
  - (3) 9 + 0 array
  - (4) 4 + 4 array
103. Choose the **correct** option for solution :-
- (1)  $\text{water} \propto \text{osmotic Pressure}$
  - (2)  $\text{OP} \propto \text{Solute concentration}$
  - (3)  $\text{OP} \propto \frac{1}{\text{solute concentration}}$
  - (4)  $\text{OP} \propto \text{DP}$
104. The transpiration in plants will be the lowest :-
- (1) When there is high humidity in the atmosphere
  - (2) high wind velocity
  - (3) there is excess of water in the cell
  - (4) environmental conditions are very dry
105. Which one element is the part of vitamins Biotene and thiamine :-
- (1) Ca
  - (2) S
  - (3) Mo
  - (4) B
106. Where are the protons and  $\text{O}_2$  Released during photosynthesis :-
- (1) Lumen of thylakoid
  - (2) In thylakoid membrane
  - (3) In stroma
  - (4) Periplastodial space
107. Which process of calvin cycle require RuBP ?
- (1) Carboxylation
  - (2) Reduction
  - (3) Regeneration
  - (4) Phosphorylation
108. Amino acid not enters in respiratory pathway at :-
- (1) Krebs cycle
  - (2) Pyruvic acid
  - (3) Acetyl Co-A
  - (4) Fructose 6 Phosphate
109. RQ of fats substances is generally :-
- (1) unity
  - (2) less than one
  - (3) greater than one
  - (4) zero
110. Which effect of auxin is applied world wide for plant Propagation :-
- (1) Apical Dominance
  - (2) Herbicidal action
  - (3) Initiate rooting in stem cutting
  - (4) Fruit ripening
111. Read the following sentences :-
- (a) In elution the separated bands of DNA are cut out from agarose gel and extracted from the gel piece.
  - (b) *E. Coli* cloning vector pBR-322 have several restriction sites, ori, antibiotic resistance genes and RoP.
  - (c) The downstream processing and quality control testing vary from product to product.
  - (d) Competent bacterial cells take up the plasmid.
- (1) All are incorrect
  - (2) Except d, all are correct
  - (3) Only d is correct
  - (4) All are correct
112. Which of the following cross with respect to blood groups show both multiple allelism and co-dominance in offsprings :-
- (1)  $B \times B$
  - (2)  $A \times A$
  - (3)  $A \times B$
  - (4)  $AB \times O$
113. The distillation process is required in the formation of :-
- (1) Rum
  - (2) Brandy
  - (3) Whisky
  - (4) All of these
114. A child with blood group genotype  $I^A I^B$  is both of a woman with genotype  $I^B I^B$ . The father could not be a man of genotype :-
- (1)  $I^A I^A$
  - (2)  $I^B I^B$
  - (3)  $I^A I^B$
  - (4)  $I^A i$

- 115.** In a vector which of the following is helpful in selection of transformants.

(1) Restriction site (2) Ori site  
(3) Selectable marker (4) Rop site

**116.** Two genes A and B present on same chromosome show 30% crossing over, then what will be the percentage of gamete (aB) for the plant having genotype Aa Bb in Cis arrangement :-

(1) 40% (2) 70%  
(3) 20% (4) 15%

**117.** In trihybrid, when all genes show complete linkage then what is ratio of test cross :-

(1) 1 : 1 : 1 : 1 : 1 : 1 : 1 : 1 : 1  
(2) 1 : 1  
(3) 1 : 1 : 1 : 1  
(4) 9 : 3 : 3 : 1

**118.** DNA finger printing profile will be exactly same of:-

(a) Sibling (b) Monozygotic twins  
(c) Offsprings (d) Dizygotic twins

(1) a and b only (2) b and c only  
(3) b only (4) b and d only

**119.** In some viruses flow of information is in reverse direction, that is, from RNA to DNA, this process is known as :

(1) Transcription  
(2) Reverse transcription  
(3) Replication  
(4) Reverse replication

**120.** The packaging of chromatin at higher level requires additional set of proteins that are collectively are referred to as :

(1) Histone protein  
(2) Non-histone chromosomal protein  
(3) Packaging protein  
(4) All of these

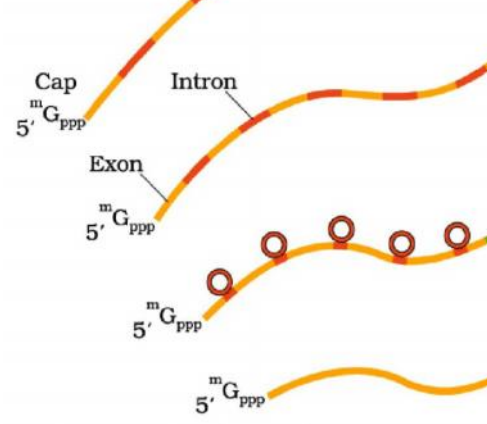
**121.** Which was the first genetic material ?

(1) DNA (2) RNA  
(3) Protein (4) Carbohydrate

**122.** There is a definite region in E.Coli DNA where the replication originates. Such regions are termed as :

(1) Origin of replication  
(2) Replication fork  
(3) Replicon  
(4) Initiation point

**123.** Below figure represents :



(1) Capping  
(2) Polyadenylation  
(3) Splicing  
(4) hnRNA processing

**124.** Hybrid vigour is due to :

(1) Chiasma (2) Inbreeding  
(3) Recombination (4) Heterozygosity

**125.** Somaclonal variation appears in :

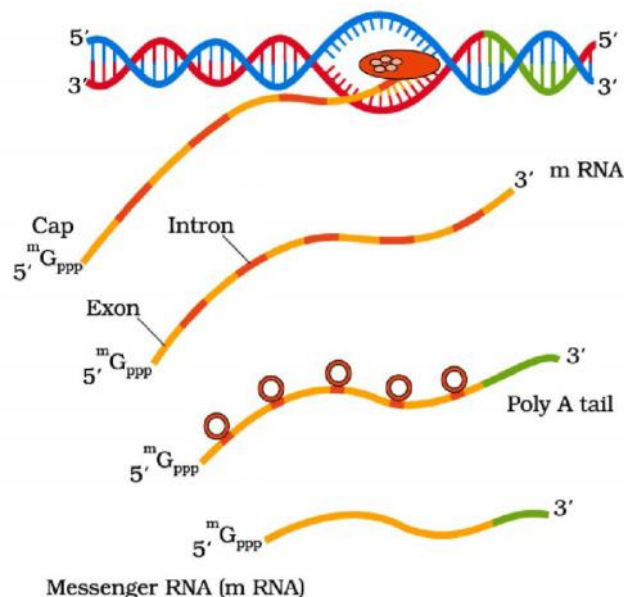
(1) Organism produced through somatic hybridization  
(2) Plants growing in highly polluted conditions  
(3) Apomictic plants  
(4) Tissue culture raised plants

**126.** Which one of the following is not a constituent of cell membrane ?

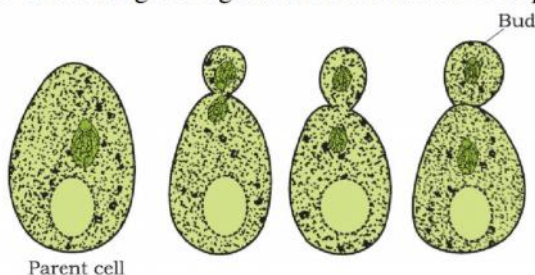
(1) Phospholipid (2) Cholesterol  
(3) Glycolipid (4) Proline

**127.** Which of the following statement **incorrect** about enzyme activity ?

(1) Each enzyme shows its highest activity at a particular temperature and pH called the optimum temperature and optimum pH  
(2) Activity declines both below and above the optimum value  
(3) Low temperature preserves the enzymes in a permanently inactive stage  
(4) High temperature destroys enzymatic activity because proteins are denatured by heat



128. Taenia is endoparasite by which method it will get its nutrition in vertebrates alimentary canal:-  
 (1) By hooks  
 (2) By suckers  
 (3) Through general body surface  
 (4) Both 1 and 2
129. Alimentary canal is complete with a well developed Muscular pharynx is present in :-  
 (1) Taenia (2) Ascaris  
 (3) Fasciola (4) Both 1 and 3
130. Members of which sub phylum shows "Retgressive metamorphosis" :-  
 (1) Protochordata (2) Urochordata  
 (3) Cephalochordata (4) Euchordata
131. What type of vision is found in cockroach ?  
 (1) Mosaic (2) Superposition  
 (3) Binocular (4) None of them
132. Simple & compound epithelium can be distinguish by ?  
 (1) No. of cells layer  
 (2) Type shape of cells  
 (3) Origin of cells  
 (4) Through plasma membrane modification
133. Mark the odd one in this series :-  
 Areolar tissue, blood, Neuron, Tendon  
 (1) Areolar tissue (2) Blood  
 (3) Neuron (4) Tendon
134. How many statements are correct from following statements (A-D) ?  
 (A) Cartilage and bone are both structural materials  
 (B) Blood is a fluid tissue with transport functions  
 (C) Areolar tissue is a reservoir of stored energy  
 (D) Muscle tissue, which can shorten in response to stimulation  
 (1) Three (2) Two (3) One (4) Four
135. Refer the given figures and select the correct option



- (1) It is a type of parthenogenesis  
 (2) It is a type of asexual reproduction  
 (3) The offsprings can also be called as clones  
 (4) Both (2) and (3)

136. Triple fusion in capsella *Bursa pastoris* is fusion of male gamete with :  
 (1) Antipodal (2) Synergid  
 (3) Secondary nucleus (4) Egg
137. Coleoptile and coleorhiza are the protective sheaths covering \_\_\_\_\_ and \_\_\_\_\_ respectively.  
 (1) Radicle, Hypocotyl (2) Radicle, Plumule  
 (3) Plumule, Epicotyl (4) Plumule, Radicle
138. Match the following & Choose right option :-

|       | Column-I       |     | Column-II  |
|-------|----------------|-----|------------|
| (I)   | Tap root       | (A) | Banyan     |
| (II)  | Stilt root     | (B) | Rhizophora |
| (III) | Prop root      | (C) | Turnip     |
| (IV)  | Pneumatophores | (D) | Sugar cane |

Option :-

- (1) I-C, II-A, III-D, IV-B  
 (2) I-C, II-D, III-A, IV-B  
 (3) I-C, II-D, III-B, IV-A  
 (4) I-A, II-C, III-B, IV-D
139. How many plants belongs to family Solanaceae?  
 Ashwagandha, Lily, Tulip, Tobacco, Belladonna, Chilli, Lupin, Tomato, Gram  
 (1) Seven (2) Four  
 (3) Eight (4) Five
140. Which of the following is not produced by the process of differentiation ?  
 (1) Primary xylem (2) Parenchyma  
 (3) Secondary xylem (4) Primary phloem
141. Bryophytes are :-  
 (1) Vascular cryptogams  
 (2) Non-Vascular cryptogams  
 (3) First successful land plants  
 (4) Vascular embryophytes
142. In Pteridophytes reduction division happens in :-  
 (1) Prothallus (2) Gametangia  
 (3) Spore mother cells (4) Zygote
143. Choose incorrect option :

|     |                 |              |
|-----|-----------------|--------------|
| (1) | Chemical method | Today        |
| (2) | Barrier method  | Diaphragms   |
| (3) | I.U.D.          | Progestasert |
| (4) | Hormonal method | CuT          |

144. The pregnancy test is based on detection of which hormone ?  
 (1) Progesterone (2) Inhibin  
 (3) HCG (4) Estrogen
145. Find out correct order of sperm atogenesis ?  
 (1) Sperm atogonia, Sperm atid, Primary sperm atocyte, Sperm  
 (2) Sperm atogonia, Primary sperm atocyte, Sperm atid, Sperm  
 (3) Primary sperm atocyte, Sperm atogonia, Sperm atid, Sperm  
 (4) Sperm atogonia, Primary sperm atocyte, Sperm, Sperm atid
146. Artificial system of classification classified plants on the basis of :-  
 (1) One or two morphological characters  
 (2) Phylogenetic characters only  
 (3) Gross morphology  
 (4) Chemical characters
147. Which of the following contain information on any one taxon :-  
 (1) Manual (2) Flora  
 (3) Monograph (4) Catalogue
148. Visible or well developed sex organ are found in :-  
 (1) Cryptogamae (2) Phanerogamae  
 (3) Bryophyte (4) Pteridophyte
149. Which of the following is a feature of human only ?  
 (1) Consciousness (2) Reproduction  
 (3) Metabolism (4) Self Consciousness
150. How many species are present in the following group of organism ?  
 Mangifera indica, Solanum tuberosum, Panthera leo, Panthera pardus, solanum nigrum.  
 (1) Four (2) Five  
 (3) Ten (4) Three
151. Virion is :-  
 (1) Nucleic acid of virus  
 (2) Antiviral agent  
 (3) Protein of virus  
 (4) Completely assembled virus outside host
152. Select the pair that consists of plant or animal bacterial diseases :-  
 (1) Cholera and typhoid  
 (2) Citrus canker and crown gall  
 (3) Malaria and dengue  
 (4) Both (1) and (2)

153. Siliceous frustules of diatoms being indestructible, piled up of the bottom of ocean and formed a thick bed over billion of year such a thick bed is known as :-  
 (1) Red sea  
 (2) Diatomaceous earth  
 (3) Pseudorocks  
 (4) Red tides
154. Completely developed pupa of silk worm is called:  
 (1) Nymph (2) Imago  
 (3) Instar (4) Naid
155. Select the correct statements.  
 (1) Trypsinogen  $\xrightarrow{\text{Chymotrypsin}}$  Trypsin  
 (2) Procarboxypeptidase  $\xrightarrow{\text{Pepsin}}$  carboxypeptidase  
 (3) Proamylase  $\xrightarrow{\text{Ptylin}}$  Amylase  
 (4) Pepsinogen  $\xrightarrow{\text{HCl}}$  Pepsin
156. Match column-I with column-II and choose the correct options :

|     | Column-I      |       | Column-II           |
|-----|---------------|-------|---------------------|
| (A) | Goblet cell   | (i)   | Antibacterial agent |
| (B) | Lysozyme      | (ii)  | Mucus               |
| (C) | Saliva        | (iii) | HCl                 |
| (D) | Oxyntic cells | (iv)  | Sublingual gland    |

**Option :**

- (1) A-iii, B-i, C-ii, D-iv  
 (2) A-ii, B-iii, C-iv, D-i  
 (3) A-iii, B-i, C-iv, D-ii  
 (4) A-ii, B-i, C-iv, D-iii
157. Which type of movement is common only shown by macrophages and Leucocytes in blood :-  
 (1) Amoeboid movement  
 (2) Ciliary movement  
 (3) Undulating movement  
 (4) Muscular movement

- 158.** Consider the following four statements (A–D) and select the option which includes all the correct ones only?
- Ammonia is very toxic and requires large amount of water for its excretion.
  - Terrestrial animals excrete less toxic nitrogenous waste products like urea & uric acid for water conservation.
  - Ammonia gets excreted as  $\text{NH}_4^+$  through general body surface or gill surface in fishes with the help of diffusion process.
  - Kidney play an important role in ammonotelism.
- Statement b, c & d
  - Statement a and b only
  - Statement a, b & c
  - Statement a, b and d
- 159.** Read the following statements :-
- It is a chronic disorder
  - Major cause of it, is cigarette smoking
  - It is an Allergy
  - Alveolar walls are damaged
  - Respiratory surface is decreased
- How many of the above statements are correct about **Emphysema** disorder :-
- Two
  - Three
  - Four
  - One
- 160.** Consider the following four statements (a-d) and select the option which includes all the correct :-
- Coronary Artery disease, (CAD) often referred to as Atherosclerosis
  - Heart failure means when the heart muscle is suddenly damaged by an inadequate blood supply
  - High blood pressure leads to heart diseases and also affects vital organs like brain and kidney
  - Angina occurs due to conditions that affect the blood flow
- Options :**
- Statements (b), (c) and (d)
  - Statements (a), (b)
  - Statements (b), (d)
  - Statement (a), (c) and (d)
- 161.** In a resting state, the central part of thick filament which is not overlapped by thin filament is called :-
- A-band
  - M-line
  - H-zone
  - I-band
- 162.**  $\text{CO}_2$  which is released out of tissues, what is **correct** for its transportation ?
- 70% of  $\text{CO}_2$  is transported by blood in the form of bicarbonate
  - 70% of  $\text{CO}_2$  is transported by blood in form of Carbamino haemoglobin, bicarbonate
  - 7% of  $\text{CO}_2$  is transported by blood in form of Carbamino haemoglobin
  - 20–25% of  $\text{CO}_2$  is transported by plasma in dissolved state
- 163.** The rate of production of organic matter during photosynthesis is known as :-
- GPP
  - Net primary productivity
  - Secondary productivity
  - All of the above
- 164.** Which biome is characterised by the absence of trees :-
- Taiga biome
  - Temperate conifer
  - Savanna biome
  - Tundra biome
- 165.** Each trophic level has a certain mass of living material at particular time called as the :-
- Fresh weight
  - Dry weight
  - Standing crop
  - Standing state
- 166.** Ecological pyramids are made for :-
- Food chain
  - Food web
  - Individual organism
  - All of the above
- 167.** How much of the incident solar radiation (ISR) is a photosynthetically active radiation (PAR) :-
- 1% - 5%
  - Less than 50%
  - 2% - 10%
  - More than 50%
- 168.** Which is not the part of Decomposition :-
- Fragmentation
  - Catabolism
  - Humification
  - Stratification
- 169.** Autumn colouration of leaves appear in the forest of region :-
- $40^\circ - 60^\circ$  latitude
  - $20^\circ - 40^\circ$  latitude
  - $20^\circ\text{N} - 20^\circ\text{S}$  latitude
  - $60^\circ$  above latitude

170. What will be happen if the number of organism increased at a place :-  
 (1) Predation on one another  
 (2) Mutualism  
 (3) Intra specific competition  
 (4) Inter specific competition
171. Productivity is the rate of production of biomass expressed in terms of :-  
 (a)  $\text{Kcal/m}^2/\text{yr}$  (b)  $\text{Kcal/m}^3/\text{yr}$   
 (c)  $\text{gm}^{-2}\text{yr}^{-1}$  (d)  $\text{gm}^{-1}\text{yr}^{-1}$   
 (1) Only a (2) c & b (3) a & c (4) a, c & d
172. Which one represents the range of condition of organism, the resource it utilises and its functional role in ecosystem.  
 (1) Niche (2) Microclimate  
 (3) Home range (4) Habitat
173. How many of the following statements are true:-  
 (a) Noise causes psychological and physiological disorders in humans  
 (b) Carbon dioxide and methane are commonly known as ozone depleting substance  
 (c) UV-B causes snow blindness  
 (d) Water prevention and control of pollution act has passed in 1974  
 (1) a, b & c (2) a, c & d  
 (3) b, c & d (4) c & d
174. The kind of epithelium which found in Bowman's capsule also found in :-  
 (1) PCT (2) Blood vessel  
 (3) Henle loop (4) Fallopian tube
175. Read the following statement carefully :-  
 (A) Tendon is example of Dense regular white fibrous connective tissue  
 (B) Ligament is a example of Dense regular yellow fibrous connective tissue  
 (C) Tendon is a example of Dense regular yellow fibrous connective tissue  
 (D) Ligament is a example of Dense irregular white fibrous connective tissue  
 Which option is correct ?  
 (1) A & B are correct (2) C & D are correct  
 (3) A & D are correct (4) B & C are correct
176. Nasal septum, tip of nose, costal cartilage, intervertebral disc, Epiglottis, ear pinna, ring of trachea  
 How many structures made up of hyaline cartilage?  
 (1) Two (2) Three (3) Four (4) Five

177. Which of the following matching is incorrect ?

|     |                         |                          |
|-----|-------------------------|--------------------------|
| (1) | Hyaline cartilage       | Most common cartilage    |
| (2) | Elastic cartilage       | Yellow fibrous cartilage |
| (3) | White fibrous cartilage | Hardest cartilage        |
| (4) | Calcified cartilage     | Head of femur & humerus  |

178. WHICH OF THE FOLLOWING IS CORRECT ?

- (A) PARS DISTALIS PRODUCES GH, PRL, TSH, ACTH, LH, FSH  
 (B) PARS INTERMEDIA SECRETES ONLY ONE HORMONE CALLED MELATONIN  
 (C) POSTERIOR LOBE OF PITUITARY IS ALSO CALLED AS NEUROHYPOPHYSIS OR PARS NERVOSA  
 (D) POSTERIOR PITUITARY, STORES AND RELEASES TWO HORMONES CALLED OXYTOCIN AND VASOPRESSIN  
 (1) A, B, C (2) B, C, D  
 (3) A, C, D (4) B AND C

- 179.

|   |                      |       |            |
|---|----------------------|-------|------------|
| A | Bone producing cell  | (i)   | Osteocyte  |
| B | Mature cell of bone  | (ii)  | Osteoclast |
| C | Bone destroying cell | (iii) | Osteoblast |

Match the correct & give answer ?

- (1) A-i, B-ii, C-iii (2) A-iii, B-ii, C-i  
 (3) A-iii, B-i, C-ii (4) A-ii, B-i, C-iii

180. IF A PERSON IS SUFFERING FROM DEPRESSION. IN SUCH SITUATION ANTIDEPRESSANT DRUGS ARE REQUIRED. THESE DRUGS  
 (1) STIMULATE  $\text{Na}^+$  INFLUX AT POST SYNAPTIC MEMBRANE.  
 (2) STIMULATE HYPER POLARISATION OF POST SYNAPTIC MEMBRANE.  
 (3) STIMULATE MORE BURSTING OF SYNAPTIC VESICLES IN PRE-SYNAPTIC MEMBRANE.  
 (4) INHIBIT THE ENZYME WHICH CATALYSE DEGRADATION OF NOR-ADRENALIN